

Q3 answer:

$7/8$: the denominator $8 = 2 \times 2 \times 2$ has only 2s as prime factors,
so we can make it a power of 10: $7/8 = 875/1000 = 0.875$, it terminates.

$1/3$: the denominator 3 is not 2 or 5, so it can never become a power of 10.
So $1/3 = 0.333\dots$ is non-terminating and repeating.